

The Imagination of Nature: Mythology as the Cultural Recording Layer of Perceptual Coupling

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Abstract

The dominant strands of cross-cultural mythography treat the symbolic content of myth as largely a function of mind and culture, with the biome supplying raw material but not structural form. An older counter-tradition, running from Schelling through Bateson, the ecological anthropology of Descola and Ingold, and contemporary 4E cognitive science, treats imagination instead as a coupling phenomenon in which perceiver and environment co-produce symbolic content; on this reading, the mythologies of cultures inhabiting different biomes should carry a recoverable visual signature of those biomes. We test whether such a signature survives in the statistical structure of anonymised mythological text by aligning Berezkin’s catalogue of folk-motifs from nearly a thousand traditions (Russian-language abstracts, geocoded to the WWF biome classification) with two image corpora, a large naturalistic sample from iNaturalist and a curated landscape set from Places365, via the SigLIP-2 vision–language model. We anonymise the motif text with an LLM-mediated pipeline that strips species mentions, place and ethnonym labels, and biome-coded vocabulary, and we stratify the image side within iconic–taxon to remove biome-by-taxon co-occurrence as a possible explanation. Anonymised motif text nevertheless retains a biome-correlated statistical signature detectable across four vision–language model architectures, with marginal $\mu\Delta$ between $+0.13$ and $+0.43 \times 10^{-3}$ depending on the model, rising to $+0.48 \times 10^{-3}$ under within-iconic–taxon stratification on the SigLIP-2 corpus, and the alignment is graded by motif breadth (strongest in biome-specific motifs and attenuating in motifs shared widely). The signal is absent in matched-format non-mythological text and, under permutation tests blocked by Glottolog macro-area (the standard grouping unit in cross-linguistic research), attenuates by roughly half but the same seven biomes that drive the headline result remain significant at $p < .05$, indicating that geographic and cultural autocorrelation contribute to but do not exhaust the effect. It survives word-order shuffling within motifs, indicating dependence on lexical content rather than sentence-level semantics, and is of comparable magnitude in a vision-blind text-to-text setting where motif embeddings are compared to biome-name targets, indicating that part of the signal reflects the embedding model’s learned text-text associations between motif vocabulary and biome words. The result is consistent with the hypothesis that ecological structure contributes to the stabilisation of mythological motifs, while leaving open the relative contributions of perceptual morphology, subsistence and activity schemas, and cultural transmission of biome-correlated style.

Significance Statement

Mythologies of cultures inhabiting different biomes carry a recoverable visual signature of those biomes, even after every species name, place name, and biome word is stripped from the text. The signature is strongest in biome-specific myths and attenuates progressively as motifs propagate across more biomes. The pattern is consistent with the hypothesis that the ecological structure of a biome contributes to the stabilisation of its mythological motifs, and reopens, in empirically tractable form, a two-century-old debate over whether imagination is a private projection of cultural meaning onto a passive world or a coupling phenomenon between perceiver and environment.

Introduction

Modern thought has long treated nature as a passive backdrop onto which the human mind projects meaning. Mythology, on this view, is what humans *do to* a world, a projection of cultural categories, anxieties, and social structures onto matter. The biome a culture inhabits supplies raw material but not, on this reading, the *form* of its symbols. We call this dominant view the *projection account*. Running parallel to this default, however, a different tradition has insisted on a different picture. Schelling’s distinction between *natura naturata* (nature as finished product) and *natura naturans* (nature as productive activity) [Schelling, 1797, 1799] was recovered in the twentieth century by Bateson’s claim that “mind is wherever the conditions for a mental process obtain” across coupled systems, where differences make differences and pattern propagates between brain, body, environment, and culture [Bateson, 1972, 1979]. Depth psychology made a related move. Jung’s mature view of the archetypes resisted easy translation into innate mental contents stored in individual brains and instead located them in the joint between psyche and nature [Jung, 1959]; Hillman radicalised this into archetypal psychology and the recovery of *anima mundi* [Hillman, 1975, 1981]; Henri Corbin gave it its most precise philosophical name in the *mundus imaginalis*, the imaginal world, distinct from the modern collapse of “imaginary” into “merely subjective” [Corbin, 1972, Barfield, 1957, Abram, 1996, Ingold, 2000]. The thread connecting these authors is that imagination is a perceptual organ rather than a productive one, an organ that discloses content present in the encounter between an attentive perceiver and an articulate world. We call this opposing view the *coupling account*, since on this reading perceiver and environment co-produce symbolic content together.

Since the 1960s the cognitive sciences have been producing the empirical scaffolding for exactly this coupling view. Gibson’s ecological psychology replaced the dominant representational model of perception with one of *direct pickup*, in which the environment is information-rich at the level of the optic array and the organism tunes itself to a richly structured world that already contains the relevant information [Gibson, 1979]. The 4E framework (embodied, embedded, extended, and enactive cognition) generalised this beyond perception [Varela et al., 1991, Clark and Chalmers, 1998, Noë, 2004]. Predictive processing supplied the neural substrate, with the brain treated as a hierarchical prediction machine that locks onto interpretations its priors privilege, given the actual statistical structure of what the world presents [Friston, 2010, Clark, 2016, Hohwy, 2013]. On this account pareidolia is not a mistake of perception but the system working as designed. Niche construction theory closed the loop [Odling-Smee et al., 2003], observing that humans produce symbolic environments (names, stories, transmitted practices) and that these environments shape the cognition of subsequent generations.

The two accounts make different empirical predictions about mythology. The projection account treats the symbolic content of myth as largely a product of mind and culture, and therefore predicts that once the text of a mythology has been stripped of its overt environmental and cultural anchors (species names, place names, ethnonyms, biome words, climate phrases, tradition labels), the residual narrative should carry no detectable visual signature of the biome the tradition inhabits. On this reading, what remains of mythology after anonymisation is the cross-cultural grammar of plot and relation, and that grammar should be indifferent to which biome a tradition emerged in. The coupling account treats imagination instead as a perceptual organ jointly tuned by mind and environment, and therefore predicts that the structural character of a biome, its light, its dominant forms, its taxa, the affordances it presents to a perceiver, should leave a recoverable residue in the surviving text, because that residue was part of what stabilised the mythology in the first place. A secondary prediction of the coupling account, which the projection account does not share, is that this residue should be strongest in motifs whose distribution remained close to a single biome and attenuate as motifs propagate across many biomes, since universally-shared motifs are precisely those whose stabilising encounters were not tied to one environment.

Two recent developments make this question newly testable at scale. The first is the cross-cultural folk-motif catalogue assembled by Yu. E. Berezkin, comprising 2,564 motifs across 958 traditions with Russian-language abstracts and geocoded centroids [Berezkin, 2024], which we have joined to the WWF Ecoregions classification of 14 terrestrial biomes [Olson et al., 2001]. The second is the development

of frontier vision–language models such as SigLIP-2 [Tschannen et al., 2025], which embed text and image in a shared metric space learned from billions of image–text pairs sampled from the internet. The combination lets us ask whether, after stripping every species name, every place name, every biome word, and every tradition label, the embedding of a tradition’s mythology preferentially aligns with images of its own biome, and whether that alignment is concentrated in biome-specific motifs as the coupling account predicts.

We report below that such a residue is detectable. After stripping every overt environmental and cultural anchor from the text, the embedding of a tradition’s mythology preferentially aligns with images of its own biome on two independent image corpora and across four vision–language model architectures, in both Russian and English. The alignment is graded with motif breadth as the coupling account predicts, strongest in motifs that did not propagate beyond a single biome and attenuating in motifs shared across many. We take this as evidence consistent with the coupling account and not naturally predicted by the projection account, and read mythology as the surviving textual residue of an ongoing perceptual coupling between cognition and environment, the recoverable trace in transmitted narrative of a living engagement that the narrative itself does not exhaust.

Results

Biome-specific mythology aligns with biome imagery

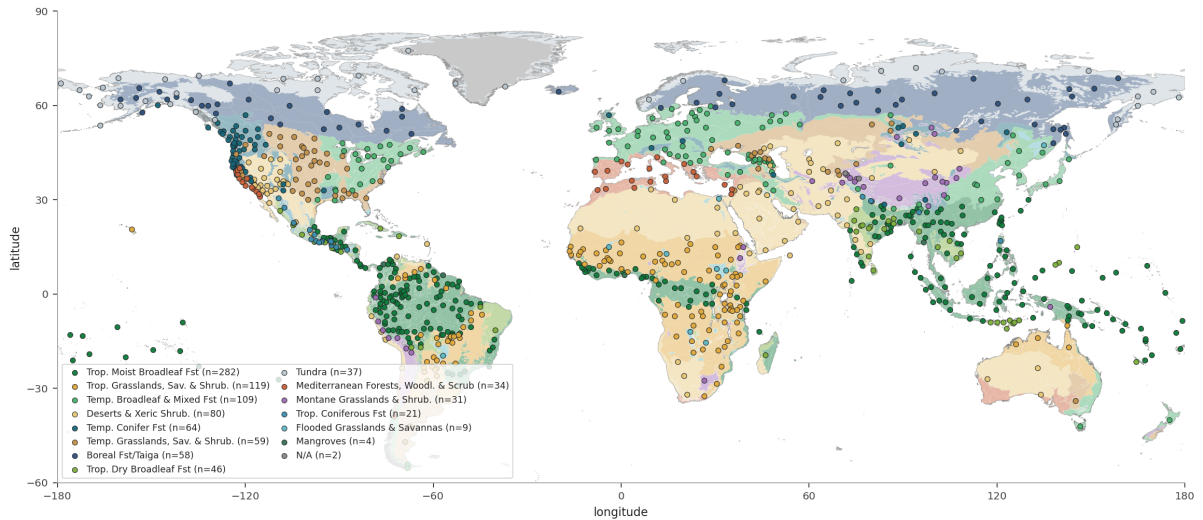


Figure 1. Cross-cultural sample of mythology over the world’s biome geography. World map of the 958 Berezkin traditions plotted as dots over the underlying WWF Ecoregions polygons, both coloured by the same biome palette: continents in semi-transparent biome fills (the Saharan and Australian deserts in tan, the Amazonian and Indo-Malay rainforests in dark green, the Boreal belt across northern Eurasia and Canada in muted blue, Mediterranean basins in red, etc.), tradition centroids as opaque dots in the same palette. The overlay shows how densely each biome’s actual extent is sampled by Berezkin’s corpus. Eurasian and North American traditions are relatively dense; some biomes (Mangroves, Flooded Grasslands and Savannas) are sparsely covered.

The Berezkin corpus distributes 958 traditions across the 14 WWF biomes (Figure 1). After the full anonymisation pipeline described in *Materials and Methods*, the LLM-mediated translation and double-pass refinement that replaces every species with a class word, every place and ethnonym with a placeholder, and every tradition-specific character name and cultural-religious title with a neutral generic, followed by a model-predictability biome-tell filter and a within-iconic-taxon stratification on the image side – the residualised effect on the full LLM-clean motif corpus and iNaturalist images reaches $\mu\Delta = +0.48 \times 10^{-3}$ stratified across the 14 biomes, with seven biomes robustly significant at $p < 0.05$. Three biomes are significant at $p < 0.001$: Montane Grasslands & Shrublands ($\Delta = +1.40 \times 10^{-3}$),

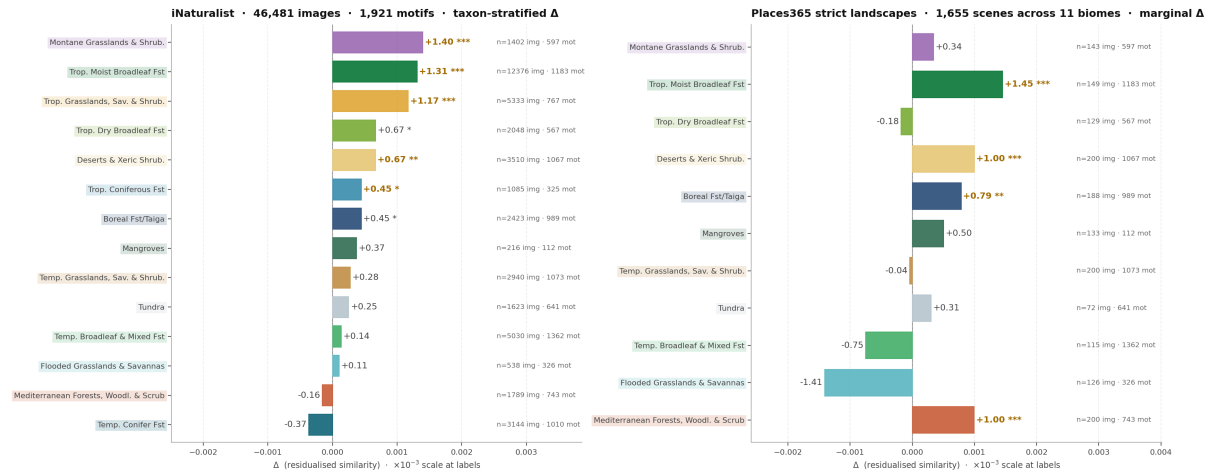


Figure 2. The headline coupling signature. Residualised Δ per WWF biome on two independent image corpora, using the full LLM-clean motif corpus after the SigLIP biome-tell filter. *Left:* iNaturalist (filtered, $n = 46,481$ images; 1,921 motifs), with Δ stratified within each iconic iNaturalist taxon to remove biome by taxon co-occurrence as a possible explanation. *Right:* Places365 strict landscapes (11 biomes), with marginal Δ (Places365 carries no iconic-taxon labels). Bars are signed Δ between motifs assigned to the biome and motifs assigned elsewhere; margin numbers show $(n_{\text{img}}, n_{\text{motif}})$ per cell; bold asterisks denote permutation-test significance ($p < .001$: ***, $p < .01$: **, $p < .05$: *). Tropical Moist Broadleaf, Tropical Grasslands & Savannas, Montane Grasslands, and Tropical Dry Broadleaf are robustly positive on both corpora; Mediterranean Forests, Deserts, and Boreal/Taiga add additional significant positive cells on at least one corpus.

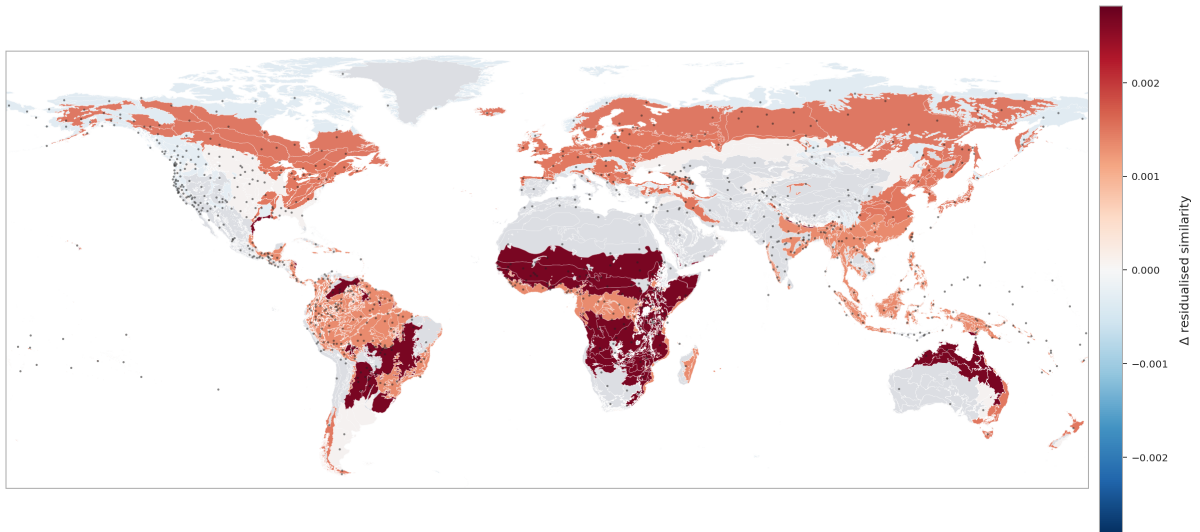


Figure 3. Per-biome Δ projected onto the world map. Geographic companion to Figure 2; each WWF biome is coloured by its iNaturalist taxon-stratified Δ on the full LLM-clean corpus. Warm colours mark biomes whose mythology aligns most strongly with their imagery (Montane Grasslands of the high-altitude tropics, Tropical Moist Broadleaf forests of the Amazon and Indo-Malay, Tropical Grasslands of Sub-Saharan Africa and Latin America, Tropical Dry Broadleaf of the seasonal subtropics, Deserts of North Africa and central Australia, and Boreal/Taiga forests of northern Eurasia and Canada).

Tropical & Subtropical Moist Broadleaf Forests ($+1.31 \times 10^{-3}$), and Tropical & Subtropical Grasslands & Savannas ($+1.17 \times 10^{-3}$). Four more biomes are significant at $p < 0.05$: Tropical & Subtropical Dry Broadleaf Forests ($+0.67 \times 10^{-3}$, $p = 0.022$), Deserts & Xeric Shrublands ($+0.67 \times 10^{-3}$, $p = 0.004$), Tropical & Subtropical Coniferous Forests ($+0.45 \times 10^{-3}$, $p = 0.017$), and Boreal Forests/Taiga ($+0.45 \times 10^{-3}$, $p = 0.027$). The remaining seven biomes, temperate forests, temperate grasslands, tundra, mediterranean forests, mangroves, and flooded grasslands, are positive or near

zero in stratified Δ but do not reach significance under the within-iconic-taxon control (Figure 2, left). The seven robustly-positive biomes span tropical, montane, desert, boreal, and coniferous habitats, ecologically diverse but unified by visually distinctive structural signatures (canopy architecture, exposed rock/sand, conifer architecture, snow line) that are evidently encoded by the iNaturalist imagery of those biomes and recovered by the mythology of cultures inhabiting them.

Mediterranean Forests is the cleanest illustration of why the within-iconic-taxon stratification matters. Its marginal Δ on the biome-specific Spec A subset reaches $+2.97 \times 10^{-3}$ ($p < .001$) and is the largest cell in the pooled per-taxon decomposition (Figure 6, Panel A, “all” column). But the per-taxon row reveals what is driving it. The Plantae cell reaches $+9.20 \times 10^{-3}$ while every other taxon for which the cell is testable, Aves (-4.08), Mammalia (-2.73), Reptilia (-3.74), Amphibia (-5.24), is sharply *negative*. This is the textbook signature of biome-by-iconic-taxon co-occurrence: Mediterranean iNaturalist imagery is dominated by plants (vineyards, olive groves, shrubland), and the biome-specific Mediterranean motifs in the catalogue are similarly plant-leaning (orchard, fig, vine, olive narratives), so SigLIP rewards “plant-text matches plant-image” regardless of whether the imagery carries any Mediterranean-specific visual character. Once we hold iconic-taxon fixed on the image side, the Plantae contribution is averaged against the negative bird and mammal contributions and Mediterranean’s Δ collapses to -0.16×10^{-3} in Figure 2. The seven biomes that survive the stratification therefore carry alignment that is *not* explainable by biome-by-taxon co-occurrence, and the biomes that collapse (Mediterranean Forests, Mangroves, Temperate Broadleaf) tell us where the marginal effect was riding on it. On the curated Places365 landscape corpus, 1,655 strict- landscape scenes across 11 biomes, drawn from Places365 scene categories that unambiguously match a single WWF biome and then SigLIP-filtered against humans, vehicles, structures, and close-up animal subjects, the marginal effect (Places365 carries no iconic-taxon labels) is on a comparable scale to the iNaturalist stratified effect ($\mu\Delta = +0.27 \times 10^{-3}$ across 11 biomes; Tropical & Subtropical Moist Broadleaf $\Delta = +1.45 \times 10^{-3***}$, Deserts & Xeric Shrublands $+1.00 \times 10^{-3***}$, Mediterranean Forests $+1.00 \times 10^{-3***}$, Boreal Forests/Taiga $+0.79 \times 10^{-3**}$; Figure 2, right). The two corpora agree on the biomes that drive the effect (tropical forest interiors, deserts, Mediterranean scrubland, boreal conifer landscape), strengthening the claim that the alignment tracks biome-character imagery and not a single corpus’s peculiarities. Projected onto a world map (Figure 3), the strongest alignments concentrate in the Amazonian and Indo-Malay Tropical Moist Broadleaf rainforests, the Saharan and Australian deserts, the Mediterranean basins of southern Europe and southwestern North America, and the boreal forests of northern Eurasia and Canada.

What the residual signal is and is not

Because the absolute magnitude of $\mu\Delta$ is small, it matters to bound where the signal comes from. We report four robustness analyses in Figure 4. First (panel A, *Encyclopedic null*), embedding random encyclopedic-register English paragraphs in place of the LLM-clean motifs and applying the same residualised test yields $\mu\Delta = +0.02 \times 10^{-3}$, essentially zero, confirming that the test statistic does not produce a positive bias on biome-neutral text. Second (panel A, *Word-shuffled motif*), shuffling the words inside each motif before re-embedding retains $\mu\Delta = +0.38 \times 10^{-3}$, about 80% of the headline value, indicating that the alignment depends on lexical content rather than sentence-level semantics: SigLIP-2 is recovering a bag-of-words signal more than a narrative-structural one. Third (panel A, *Block-permutation null*), permuting motif-to- biome assignments only within blocks that preserve geographic and cultural autocorrelation drops $\mu\Delta$ to $+0.21 \times 10^{-3}$, roughly half the headline. We construct the block in two ways. The first blocks by WWF biogeographic realm (Palearctic, Indomalayan, Neotropic, Nearctic, Afrotropic, Australasia, Oceania); under this null three biomes survive at $p < .01$, but three additional biomes (Mediterranean Forests, Montane Grasslands, Deserts) cannot be tested because each is the only biome of its realm in our corpus, which collapses their block to the observed value. The second blocks by Glottolog macro-area (Africa, Australia, Eurasia, North America, Papunesia, South America), the standard grouping unit in cross-linguistic research, and gives a real block test for all 14 biomes; under this null seven biomes survive at $p < .05$ (Tropical & Subtropical Moist Broadleaf $+0.80 \times 10^{-3}$, $p < .001$; Deserts & Xeric Shrublands $+0.49 \times 10^{-3}$, $p < .001$; Mediterranean Forests $+0.51 \times 10^{-3}$,

$p = .001$; Temperate Broadleaf & Mixed $+0.51 \times 10^{-3}$, $p = .002$; Montane Grasslands & Shrublands $+0.48 \times 10^{-3}$, $p = .004$; Tropical & Subtropical Grasslands and Savannas $+0.83 \times 10^{-3}$, $p = .015$; Tropical & Subtropical Dry Broadleaf $+0.25 \times 10^{-3}$, $p = .031$), the same seven biomes that drive the headline result. Geographic and cultural autocorrelation therefore contribute to but do not exhaust the effect. Fourth (panel B), training binary classifiers to predict each biome from the SigLIP-2 text embedding alone yields a macro AUROC between 0.57 and 0.63 across all anonymisation stages (raw, hypernym v3 through v8, and the LLM-clean pipeline), modestly but consistently above the chance value of 0.5, confirming that biome-correlated lexical structure persists in the text even after our strictest anonymisation. A separate vision-blind probe (panel A is one such control; the parallel text-to-text version replaces image prototypes with biome-name prompt embeddings) gives a $\mu\Delta$ of comparable magnitude to the visual prototype version, which indicates that part of the alignment we measure with images reflects the embedding model’s learned text-to-text association between motif vocabulary and biome-name vocabulary rather than a purely visual coupling. We return to the implications of this in the Discussion.

A second concern, raised both by the audit and by the structure of our anonymisation rubric, is that the pipeline targets nouns (species, places, biomes, ethnonyms) but retains by design a wide class of generic landscape words (forest, mountain, river, snow, sand) and activity verbs (gather, hunt, fish, herd) that are themselves heavily biome-correlated. An expanded audit of 200 random motifs (panel C) confirms that explicit biome words appear in 0% of cleaned motifs, place or ethnonym residues in 0.5%, and specific species references in 3.0% (after excluding verb-ambiguous matches such as “to swallow” or “to bear” flagged by a sloppy keyword search), giving a composite strict-clean rate against species, place, and biome vocabulary of 96.5% across the sample. The two remaining residue categories, generic landform vocabulary (92.5% of motifs) and biome-correlated activity verbs (93.5%), are retained by construction; in the Discussion we treat them as part of the content over which the alignment is measured rather than as contamination to be removed. Cross-model robustness on the LLM-clean corpus (panel D) confirms that the alignment is recovered in all four vision–language model families tested. Under the marginal definition applied uniformly across all four models, $\mu\Delta$ ranges from $+0.13 \times 10^{-3}$ on OpenCLIP-ViT-L/14-OpenAI to $+0.43 \times 10^{-3}$ on OpenCLIP-ViT-L/14-LAION-2B, with SigLIP-2 at $+0.31 \times 10^{-3}$ and M-CLIP at $+0.23 \times 10^{-3}$. Under the headline within-iconic-taxon stratification on SigLIP-2, $\mu\Delta$ rises to $+0.48 \times 10^{-3}$, the value reported in panel A and used throughout the rest of the paper. All four models give a positive mean Δ on the same anonymised text, ruling out the strongest single-model artefact reading.

The coupling-versus-projection discriminating test

The single test that most sharply distinguishes the two accounts is the breadth-stratified analysis in Figure 5. Under the projection reading we would expect roughly similar per-biome alignment magnitudes for biome-specific and universal motifs, since both are equally available as cultural projections of biome-correlated style. Under the coupling reading, only the biome-specific motifs are stable points of encounter with their biome’s visual structure; universals propagated precisely because they detached from any specific biome and stabilised on cross-cultural structures (family conflict, lost-then-found, transformation). The breadth-split result is decisive in the predicted direction: Spec A motifs show $\mu\Delta = +0.00139$ with 5 of 9 cells significant under FDR; semi-universal motifs show $\mu\Delta = +0.00024$ with 1 of 14 cells under FDR; universal motifs show $\mu\Delta = +0.00033$ with 0 of 14 under FDR. The signature is present where the coupling account locates it and absent where the projection account would have it equally present.

Per-taxon decomposition

The biome–imagery alignment is not localised to any single taxonomic class. The pooled all column reaches $\mu\Delta = +1.06 \times 10^{-3}$ across nine biomes. Aves and Mammalia carry the largest taxon means ($+2.61$ and $+1.60 \times 10^{-3}$), followed by Actinopterygii, Plantae, Mollusca, Animalia, and Insecta (all positive on average); Fungi, Reptilia, Arachnida, and Amphibia have weakly negative means. But every taxon, including those with negative means, has at least one significantly positive biome cell. The pattern

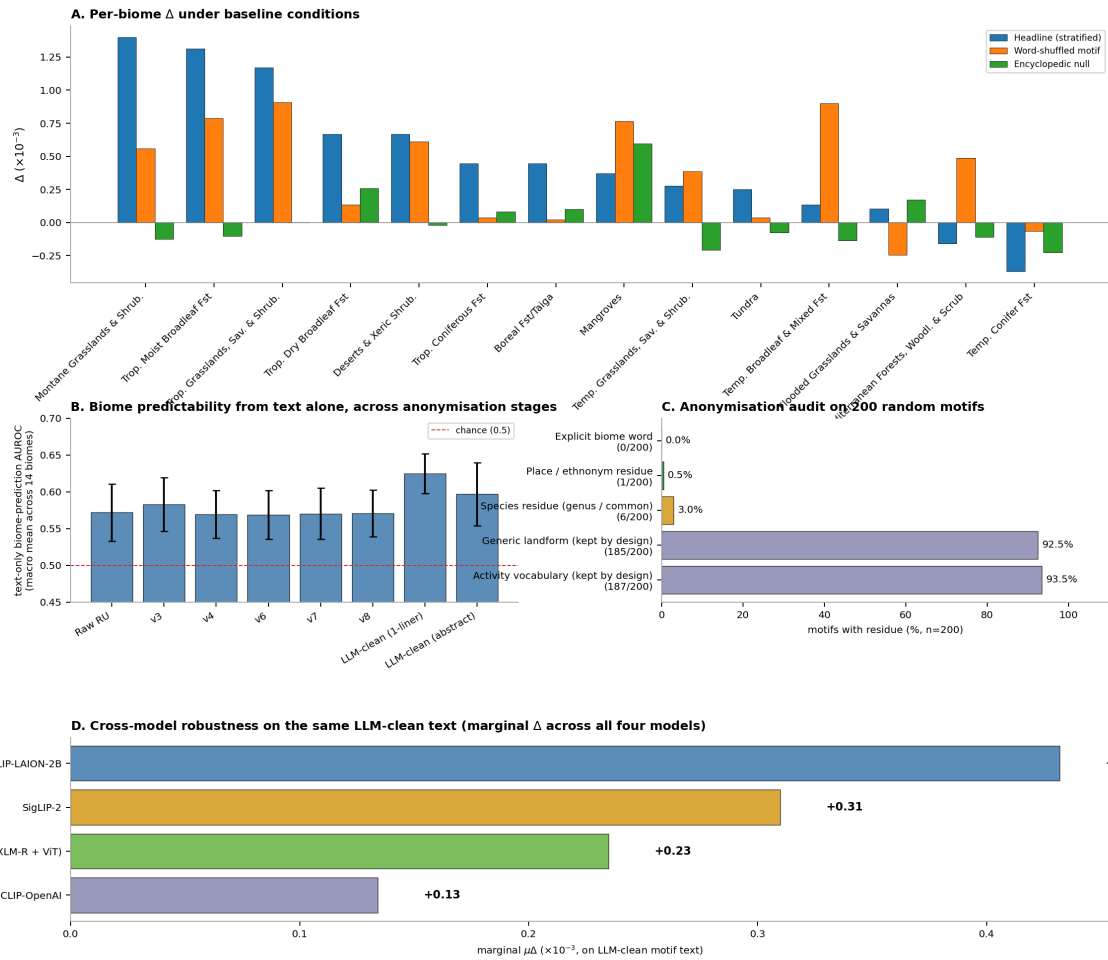


Figure 4. Robustness analyses. *Panel A:* per-biome observed Δ under three conditions that modify the input. The headline within-iconic-taxon stratification (blue) is what the rest of the paper reports. Within-motif word-shuffling (orange) destroys sentence order while preserving the vocabulary of each motif. The encyclopedic null (red) replaces motif texts with random encyclopedic-register English paragraphs of comparable length. The Glottolog macro-area block-permutation null discussed in the text shifts only the significance threshold, not the observed Δ , so it is not plotted here; under that null seven biomes (Tropical/Subtropical Moist Broadleaf, Deserts & Xeric Shrublands, Mediterranean Forests, Temperate Broadleaf & Mixed, Montane Grasslands, Tropical/Subtropical Grasslands & Savannas, Tropical/Subtropical Dry Broadleaf) remain significant at $p < .05$, the same seven biomes that drive the headline. *Panel B:* macro AUROC of per-biome binary classifiers trained on SigLIP-2 text embeddings at each anonymisation stage; chance (0.5) is the dashed red line. AUROC is 0.57–0.63 at every stage, modestly above chance and not collapsed by anonymisation. *Panel C:* expanded 200-motif audit, per-criterion residue rate. Biome words, place or ethnonym residues, and specific species references are essentially eliminated (0%, 0.5%, and 3.0% respectively, composite strict-clean rate 96.5%). Generic landform vocabulary and biome-correlated activity verbs are retained by design and are part of the content over which the alignment is measured. *Panel D:* cross-model headline $\mu\Delta$ on the same LLM-clean motif text, reported under the marginal definition uniformly across all four model architectures so the comparison is apples-to-apples. SigLIP-2’s stratified headline value of $+0.48 \times 10^{-3}$, used in the rest of the paper, can be read as the marginal SigLIP-2 value of $+0.31 \times 10^{-3}$ plus the gain from within-iconic-taxon stratification on the iNat side.

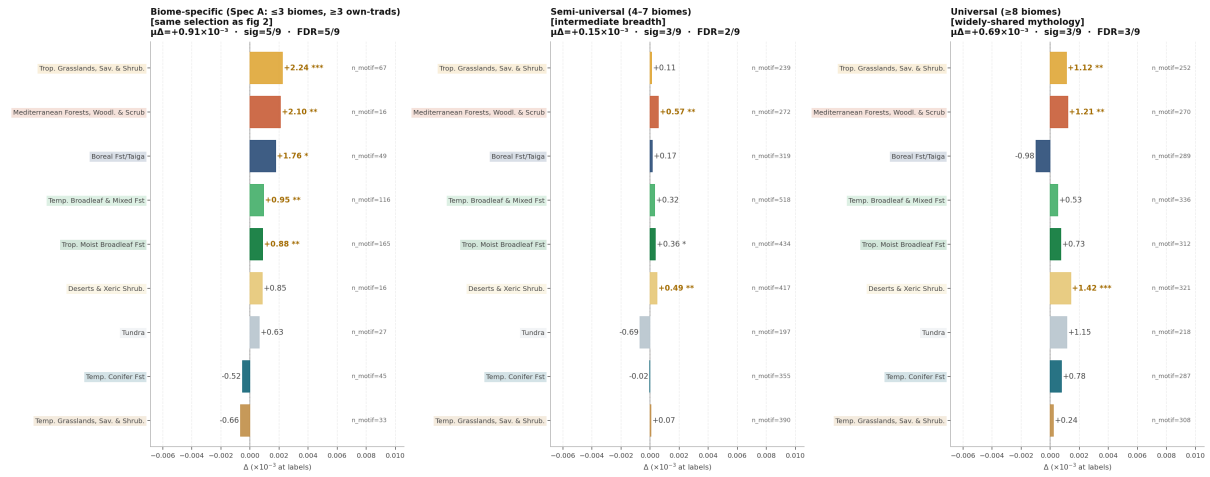


Figure 5. Breadth-stratified analysis. Three panels stratify motifs by how many biomes they touch. *Left*: Spec A motifs (≤ 3 biomes touched, ≥ 3 own-traditions per biome), the headline subset. *Centre*: semi-universal (4–7 biomes). *Right*: universal (≥ 8 biomes). The biome–imagery alignment is concentrated in the Spec A panel and absent from universals, the pattern the coupling account predicts and the projection account does not.

is ecologically interpretable. Each biome’s mythology aligns most strongly with the taxa morphologically characteristic of that biome, Aves and fish dominate the Tundra and Boreal cells, Plantae and Fungi dominate Mediterranean and Temperate Broadleaf, Reptilia and Amphibia have their strongest positive cells in Tropical Moist Broadleaf and Tropical Grasslands. The Kruskal–Wallis omnibus across the twelve taxa is not significant, and no individual taxon survives BH correction across taxa. With only nine biome cells per taxon, the test has limited power to discriminate distributions, and we do not claim a statistical separation between taxa. But the ranked pattern is consistent. Aves and Plantae sit above the pooled aggregate and above every animal class collapsed into Animalia; the cells that pass cell-level FDR concentrate in Aves and Plantae panels (Tropical Grasslands, Mediterranean, Boreal) as well as scattering across Mammalia, Amphibia, and Reptilia. A reasonable reading is that birds and plants contribute somewhat more strongly than other classes to the biome-mythology coupling, even though they are typically not the explicit *subjects* of the motifs after anonymisation. Their visual structure (silhouettes, flight, plumage colour patterns; foliage texture, branching geometry, seasonal palette) appears to carry more biome-discriminative information than the visual structure of, say, arachnids or molluscs, and the coupling we measure reflects that asymmetry. The per-taxon decomposition therefore does more than rule out a single-class artefact. Because the motif text the encoder ingests has already been anonymised, the specific species, plant, and fungus names are no longer in the text; the per-biome similarity is being computed between, for instance, biome-b photographs of *specific* birds and motif text in which every bird name has been replaced by the generic class word “bird”. The fact that the photographs of biome-b birds nonetheless align preferentially with biome-b motif text means that the encoder is recovering a visual correspondence between the actual birds of biome b and the residual narrative content of motifs that emerged in biome b, despite the motif text containing no bird identity at all. The same logic holds for plants, mammals, and fungi. The residual signal in each taxonomic panel reflects a coupling between the visual structure of that biome’s flora or fauna and the narrative residue of mythology that stabilised in encounter with it, not a vocabulary match between species names and species images. This is the strongest single indication in the data that the alignment is content-driven rather than vocabulary-driven, and it distributes across the ecological content of the mythology rather than residing in any one organismal kingdom.

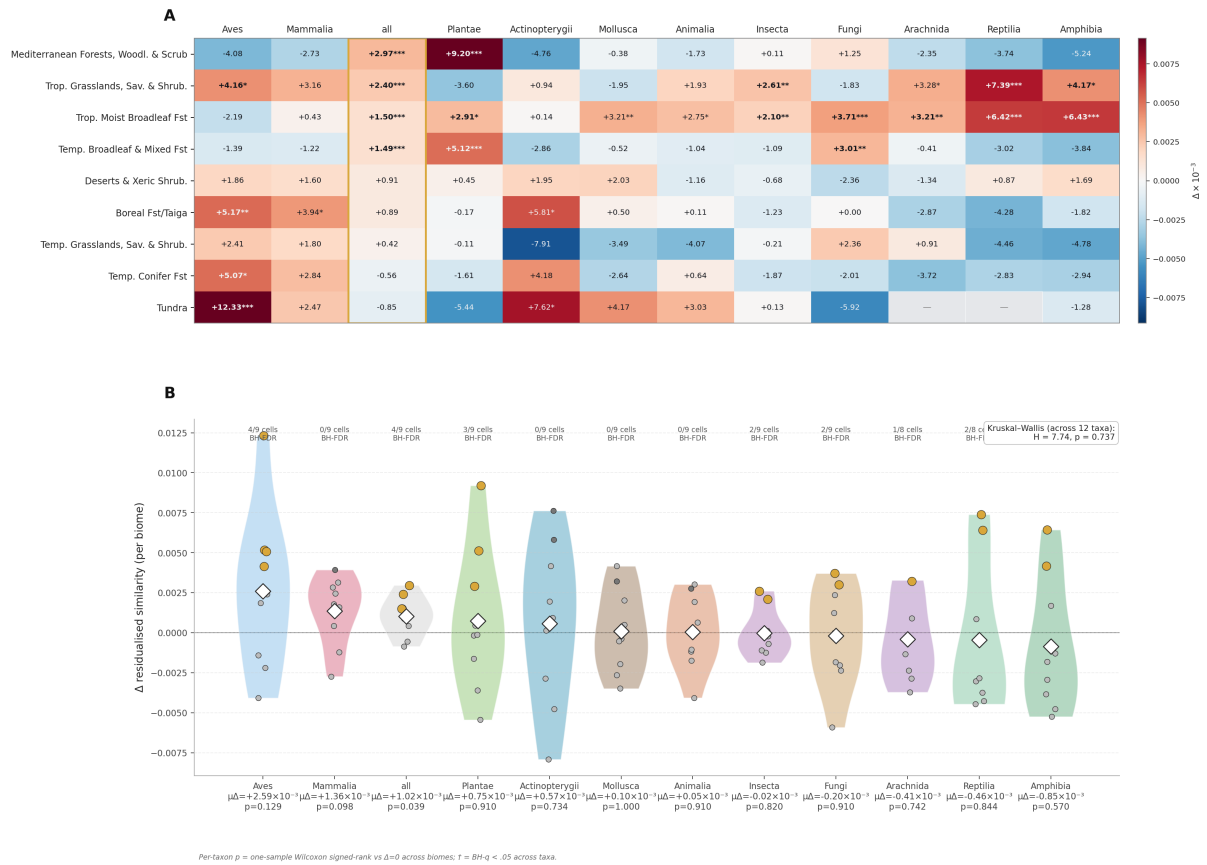


Figure 6. Per-taxon decomposition of biome alignment. *Panel A:* biome $\times$ taxon matrix of residualised Δ values (in units of 10^{-3}), with biomes ordered top-to-bottom by descending pooled (“all”) Δ and taxa ordered left-to-right by descending mean Δ across biomes. Cell values report the per-biome per-taxon Δ , with stars for nominal significance and bold for BH- $q < 0.05$ within the taxon; missing cells (below MIN_MOTIFS = 10) are shown as grey em-dashes. The gold-bordered all column anchors the headline pooled effect. *Panel B:* violin plot of the same data collapsed to per-taxon distributions. White diamonds mark the per-taxon mean; dots are individual biome cells (gold = BH-FDR significant at the cell level; grey = nominal $p < 0.05$; pale grey = n.s.). Under each violin we report the across-biomes mean Δ (in units of 10^{-3}) and a one-sample Wilcoxon signed-rank p -value testing $\Delta = 0$. The boxed annotation reports a Kruskal-Wallis omnibus across taxa. The pattern across both panels is ecologically interpretable. Each biome aligns most strongly with the taxa morphologically characteristic of that biome, and the pooled all effect ($\mu\Delta = +1.06 \times 10^{-3}$) is carried jointly by Aves, Mammalia, and the per-biome ecologically appropriate vertebrate and invertebrate taxa rather than concentrated in any single class.

Discussion

A long line of philosophical and anthropological thought has held that imagination is not a private faculty projecting cultural meaning onto a passive world but a coupling phenomenon between an attentive perceiver and an articulate environment. If this is right, then the mythologies of cultures inhabiting different biomes should carry, in their narrative structure, a recoverable visual signature of the environments that shaped them, a signature distinct from the explicit lexical content (the names of species, places, biomes, and peoples) that earlier theorists have long noted varies with geography. The empirical question we ask is whether such a signature, stripped of those lexical anchors, survives in the statistical structure of the text itself.

We find that it does, with important qualifications. Anonymised mythology in 958 traditions preferentially aligns with images of those biomes on two independent image corpora and across four vision-language model architectures, and the alignment is graded with motif breadth, strongest in motifs whose distribution remained close to a single biome and attenuating in motifs that propagated widely. The signal

is absent in matched-format non-mythological text and, under a Glottolog macro-area block-permutation null that preserves geographic and cultural autocorrelation, attenuates by roughly half but the same seven biomes that drive the headline result remain significant at $p < .05$, indicating that autocorrelation contributes to but does not exhaust the effect. The signal is preserved when motif word order is destroyed by within-motif shuffling, indicating that it depends on lexical content rather than sentence-level semantics. It is also of comparable magnitude when we replace the image side with biome-name text prompts and compare in a vision-blind setting, indicating that part of the alignment reflects the embedding model’s learned text-to-text associations between motif-content vocabulary and biome-name vocabulary, not only a text-to-image perceptual coupling. The honest summary is therefore that ecological structure leaves a recoverable statistical residue in the surviving text of mythology, and that this residue is partly lexical, partly compositional, and partly tied to the embedding model’s training distribution; what remains underdetermined by the present analysis is the relative contribution of perceptual morphology, subsistence and activity schemas, and cultural transmission of biome-correlated style. We return to this underdetermination below.

On the projection account, mythology’s symbolic content is largely a function of culture; the biome supplies raw material but not symbolic form. That account would predict that once the direct environmental references are stripped from the text (species names, place names, biome words, geographic regions, climate phrases), the residual mythology should no longer correlate with images of its biome. We strip all of that and the correlation remains. The projection account can be reconciled with this only by appealing to indirect cultural transmission of biome-correlated style, but each such reconciliation costs an auxiliary hypothesis, and the resulting account no longer makes risky predictions. Critically, the projection account also has no natural explanation for why the alignment is graded by motif breadth, strongest in biome-specific motifs, weaker in semi-universal motifs, and weakest (but not zero) in universals (Figure 5). On the coupling account, by contrast, this gradient follows directly. Biome-specific motifs are precisely the ones that stabilised in encounter with a particular biome’s visual structure, while universals stabilised on cross-cultural structures that depend only weakly on any specific environment.

The lineage that natively predicts the result runs across two centuries. Schelling’s distinction between *natura naturata* and *natura naturans* held that mind does not arise *in* nature as one more product; mind is nature considered as productive activity becoming aware of itself [Schelling, 1797]. Bateson translated this into a precise cybernetic claim, that mind is wherever the conditions for a mental process obtain, where differences make differences, where pattern propagates across coupled systems [Bateson, 1972, 1979]. A human imagining a figure in a cliff face is not generating an image from inner resources and applying it to outer matter; the imagining is an event in a coupled system of neural pattern-completion machinery, the actual statistical structure of the cliff, the cultural priors that have shaped what counts as a figure, and the linguistic apparatus that will or will not stabilise the percept into a transmissible name. Pull any element from the system and the imagining does not occur, or occurs differently. Depth psychology developed a parallel challenge to projection: Jung’s mature view located the archetypes neither in nature alone nor in psyche alone but in the joint [Jung, 1959]; Hillman held that imagination is responsive, finding figures that are there to be found in the encounter between attentive perceiver and articulate world [Hillman, 1975, 1981]; Corbin’s *mundus imaginalis* named a domain of real forms encountered through trained imagination, distinct from the modern collapse of “imaginary” into “merely subjective” [Corbin, 1972]. Beginning with Gibson’s ecological psychology [Gibson, 1979] and generalised by the 4E program [Varela et al., 1991, Clark and Chalmers, 1998, Noë, 2004], the cognitive sciences produced the empirical scaffolding for this view. Predictive processing supplies the mechanism [Friston, 2010, Clark, 2016, Hohwy, 2013]; niche construction theory closes the loop [Odling-Smee et al., 2003].

A figure in nature (the bear in the stars, the giant in the mountain, the serpent in the fjord) is on this synthesis the stable point of a three-way coupling between environmental structure (the actual statistical regularities present in the perceivable scene), perceptual machinery (priors that privilege faces, bodies, agents, predators), and cultural transmission (which stabilises particular completions, names them, carries them across generations, and binds them into narrative form). In this frame, imagination is not

the private generative act of a sealed mind; it is the moment of stabilisation in this three-way coupling. Mythology, on this reading, is the cultural recording layer of perceptual coupling between cognition and environment. What gets recorded is not what humans alone produced; it is what the environment, through human perceptual and cultural machinery, brought into articulable form.

The fact that a vision–language model (a system with no perceptual phenomenology, no cultural participation, and no priors shaped by deep evolutionary time) recovers the environment–mythology correspondence is significant for a specific reason. The correspondence is not merely a fact about how a particular human culture experienced its world. It is a fact about the joint statistical structure of environments and the mythologies that emerged within them, robust enough that a system without phenomenology can detect it. This converts the philosophical lineage from a speculative reading into a falsifiable program. If imagination is coupling, then non-conscious systems trained on the products of that coupling should recover its signature in the data; they do. This is not proof of the strongest framing, since the result remains compatible with weaker readings, but it is the kind of evidence the framing predicts and that the projection account does not.

The data do not show that nature is conscious, or that the universe has intention, or that mythologies are messages from a non-human agency. They show that the joint distribution of environments and mythologies carries structure detectable by a multimodal embedding model. The philosophical lineage offers a reading of this structure that does not collapse back into the projection model. Whether one is willing to extend the reading further, toward Whitehead’s process metaphysics [Whitehead, 1929] or toward Goff’s panpsychism or Strawson’s realistic monism [Goff, 2019, Strawson, 2006], is a separate choice the data neither compel nor forbid. The strongest defensible claim is that the empirical correspondence is consistent with, and gives new traction to, a tradition of thought in which imagination is a coupling phenomenon and mythology is its recording layer. The work makes that tradition newly testable.

Several limits should be acknowledged. Berezkin’s corpus is curated and Eurasian-weighted; some biomes (Mangroves, Flooded Grasslands, Temperate Conifer Forests) are sparsely covered and do not reach significance under the strict within-iconic-taxon control. SigLIP-2 carries its training-data biases; the cross-model replication on M-CLIP, OpenCLIP-LAION-2B, and OpenCLIP-OpenAI bounds this contribution from below but does not eliminate it.

A residual concern, which deserves explicit framing, is that the anonymisation pipeline targets nouns and noun phrases (species names, place names, biome words, geographic regions, climate descriptors) but leaves verbs and activity predicates untouched. A motif sentence such as “a woman gathers berries, steps three times in mammal dung, and is taken” contains no species name and no biome word after anonymisation, but the activity vocabulary (gathering berries, stepping in dung) is itself heavily biome-correlated in the training distribution of internet vision–language models: “gathering berries” co-occurs with temperate and boreal forest imagery, “casting fishing nets” with coastal imagery, “herding reindeer” with tundra imagery. We therefore cannot fully distinguish two readings of the residual alignment. On the *morphology* reading, the encoder is finding a visual correspondence between the actual flora, fauna, and terrain of biome b and the residual narrative content of motifs that emerged in biome b . On the *activity-priming* reading, the encoder is matching biome-correlated behaviours described in the text to biome-correlated images of the activities or their settings. Two observations are relevant. First, the projection account predicts neither. On either reading, what survives the anonymisation pipeline is a coupling between environment and culturally-stabilised narrative content. The activity vocabulary that remains is itself biome-coupled because the behaviours it describes are biome-coupled; verbs of berry-gathering and reindeer-herding survive in mythology because the activities survived in the biome. The morphology-versus-activity distinction therefore partitions the coupling account into two sub-readings; it does not rescue the projection account. Second, the Places365 result is partially informative. The Places365 strict-landscape corpus is scene-only imagery: terrain, vegetation, and climate cues, with no humans, no species, and no depictions of activities. If activity-priming alone were the mechanism, the encoder would have no purchase on Places365 imagery, since the activities described in motif text have no direct visual correlates in the photographs. Instead, the effect appears clearly on the Places365 landscape corpus (Tropical Moist Broadleaf $\Delta = +1.45 \times 10^{-3}$, Deserts & Xeric Shrublands $+1.00 \times 10^{-3}$,

Mediterranean Forests $+1.00 \times 10^{-3}$, Boreal Forests/Taiga $+0.79 \times 10^{-3}$, all $p < 0.01$, Figure 2), which means that motif content (whether morphological or activity-laden) is aligning with terrain morphology in the images. The morphology-versus-activity distinction therefore partitions the coupling account into two compatible sub-readings, biome-character morphology and habitat-bound activity, each a form of ecological coupling rather than projection. We return to disentangling them with a follow-up activity-vocabulary anonymisation pass.

Three further directions are open for follow-up work. First, the same framework should apply to mythological visual art; a corpus drawn from museum collections is under construction, and we predict that the alignment will be stronger in art than in text because the visual residue is more direct. Second, extension to recorded oral-narrative audio would test whether the coupling signature persists through a less compressed transmission channel. Third, mythologically-informed priors could be used as conditioning for scene classification. If imagination is coupling, motif-conditioned priors should help a vision system parse the scenes those motifs were stabilised in.

Schelling wrote that “nature should be visible Mind, and Mind invisible Nature.” Two centuries later, with a corpus of folk-motifs from 958 cultures, a vision–language model trained on the public internet, and an anonymisation pipeline stripped down to the residual signal, we find that mythologies of cultures inhabiting different biomes carry a recoverable visual signature of those biomes. We do not claim that the result demonstrates Schelling’s stronger metaphysical thesis. We do claim that it is the kind of evidence that thesis predicts and the modern projection account does not. The joint distribution of environments and the stories told within them is structured enough that a system without phenomenology can recover the structure, and that, we suggest, makes the coupling hypothesis a serious candidate for what mythology has been doing all along.

Materials and Methods

Corpus and biome assignment

We use Yu. E. Berezkin’s *Analytical Catalogue of World Mythology and Folklore* [Berezkin, 2024], scraped from `ruthenia.ru` in May 2024. The catalogue provides 2,564 motif entries indexed across 958 traditions; each motif has a multi-paragraph Russian-language abstract aggregating the variants found across the traditions that carry the motif. Tradition centroids are joined to WWF Terrestrial Ecoregions [Olson et al., 2001] to assign each tradition, and through it each tradition–motif pair, to one of the 14 WWF biomes.

Image corpora

The iNaturalist component consists of 46,481 images drawn from research-grade observations on iNaturalist Open Data [iNaturalist, 2024] across the 14 WWF biomes. We apply an image-side sanity filter that drops obvious non-nature content via SigLIP-2 zero-shot scoring against twelve negative prompts (“a person in the photo”, “a portrait of a human”, “people holding something”, “an angler holding fish”, “an indoor scene”, “a kitchen or laboratory”, “a city skyline or street”, “a car or vehicle”, “a road sign”, “a black or all-white image”, “a museum specimen on a white background”, “a screenshot or document”). An image is dropped if its maximum cosine similarity to any of the twelve negative prompts exceeds 0.08. The threshold is lenient (removing only obvious cases) so that species-in-habitat close-ups and natural outdoor scenes are retained, which is the content iNaturalist is biased toward by design. For the curated landscape comparison, the Places365 component [Zhou et al., 2018] is drawn from scene categories whose semantic interpretation unambiguously corresponds to a single WWF biome (e.g. *tundra* → arctic tundra; *rainforest* → tropical moist broadleaf; *forest/broadleaf* → temperate or tropical dry broadleaf; *snowfield/mountain_snowy* → boreal/taiga; *lagoon/swamp/creek/coast* → mangroves); categories that mix biomes were excluded. Each candidate image is then passed through a strict-landscape SigLIP-2 filter that drops anything that is not a wide-shot natural scene: five positive prompts (“a wide-

shot natural landscape with no people”, “a panoramic photograph of natural scenery”, “an empty natural habitat photographed from a distance”, “a landscape photo of natural terrain showing vegetation and horizon”, “a wild ecosystem with no human-made structures”) and eleven negative prompts (“a person in the photograph”, “a portrait of one or more humans”, “a close-up photo of a single animal”, “a single bird filling the frame”, “a car or vehicle”, “a boat”, “a road or street”, “a man-made structure or building”, “an indoor scene”, “a person riding a horse or animal”, “a tourist photograph with people”) are encoded by SigLIP-2 and compared against each image’s embedding. The strict-landscape score is $\text{mean}(\text{positives}) - \text{max}(\text{negatives})$; an image is dropped if its $\text{max}(\text{negatives}) > 0.18$, and the remaining top- K images per biome are kept. The Places365 filter is intentionally stricter than the iNaturalist filter, in particular it removes single-species close-ups, because the Places365 panel functions as a scene-only robustness check on biome-character imagery, complementary to iNaturalist’s species-in-habitat coverage.

Vision–language embedding

The headline model is SigLIP-2-large (google/siglip2-large-patch16-384) [Tschannen et al., 2025], a multilingual vision–language model with native Russian support. Text is tokenised with `padding=max_length`, `truncation=True`, `max_length=64`; images use the standard 384×384 preprocessing. Text and image features are L2-normalised and similarity is the inner product. For the cross-model robustness analysis we additionally embed with M-CLIP (XLM-Roberta-Large + CLIP-ViT-L/14) [Carlsson et al., 2022] on the Russian abstracts, and with OpenCLIP-ViT-L/14 (LAION-2B and OpenAI pretrainings) on English motif oneliners (see Supplementary Figure 9).

Controls against biome cues

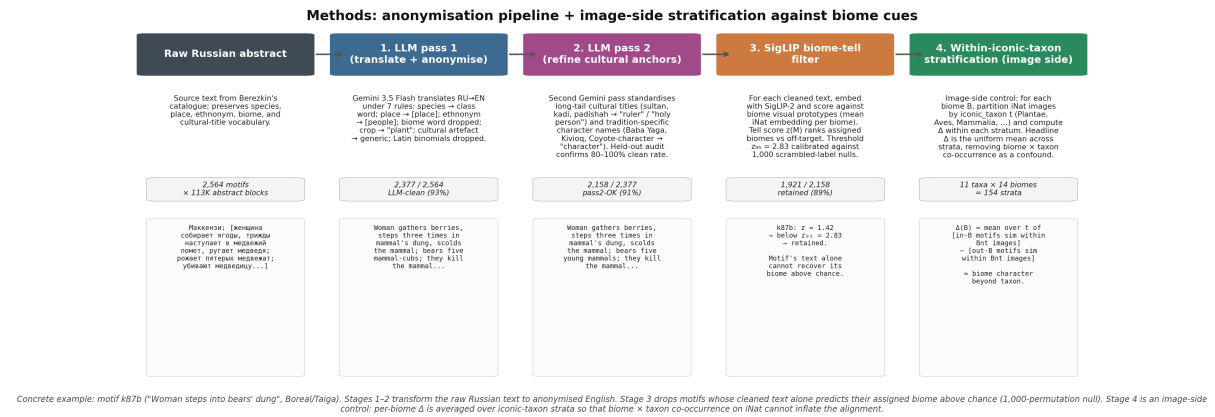


Figure 7. Anonymisation pipeline and image-side stratification against biome cues. *Stage 1.* Gemini 3.5 Flash translates each Russian Berezkin abstract into English under seven structured rules (species → class word; place → [place]; ethnonym → [people]; biome words dropped; crops → “plant”; cultural artefacts → generic; Latin binomials dropped). *Stage 2.* A second Gemini pass standardises long-tail cultural titles (sultan, kadi, padishah → “ruler” / “holy person”) and tradition-specific character names (Baba Yaga, Kivioq, Coyote-character → “character”) to neutral generics; held-out audit confirms a 80–100% clean rate. *Stage 3.* Each cleaned text is embedded by SigLIP-2 and scored against the biome visual prototypes (mean image embedding per biome). Motifs whose tell $z(M)$ exceeds the 95th percentile of a 1,000-permutation scrambled-label null are dropped ($z_{95} = 2.83$, 237/2,158 motifs excluded). *Stage 4.* The Δ test is recomputed within each iconic-taxon stratum on the image side and averaged uniformly, removing biome by taxon co-occurrence as a possible explanation of alignment. The displayed example is motif k87b (“Woman steps into bears’ dung”, Boreal/Taiga).

The motif text fed to the encoder is anonymised against every category of biome-anchoring vocabulary we could identify, by an LLM-mediated translation and refinement pass followed by a single exclusion filter. For each motif we send the raw Russian Berezkin abstract together with the original English

onelineer to gemini-3.5-flash (Google’s frontier flash model, released May 2026) [?] with a structured prompt that asks it to produce a faithful English translation under the following rules:

- Every specific species name $\rightarrow$ its class word: one of `mammal / bird / fish / reptile / amphibian / insect / mollusc / crustacean / arachnid / tree / flower / plant / fungus / animal`.
- Every country, region, river, mountain, lake, sea, ocean, or city name $\rightarrow$ the literal string `[place]`.
- Every tribe, ethnic group, nation, or ethno-religious community name $\rightarrow$ `[people]`.
- Every biome word (tundra, taiga, savanna, jungle, rainforest, prairie, desert, steppe, swamp, marsh, oasis, glacier, veldt, mangrove, mediterranean, etc.) dropped.
- Every cultivated crop (rice, corn, beans, peanuts, millet, quinoa, cassava, cotton, indigo, tobacco, etc.) $\rightarrow$ `plant`.
- Every biome-coded cultural artefact (sledge, kayak, harpoon, igloo, yurt, blowgun, snowshoes, kachina, calabash) $\rightarrow$ a generic equivalent (tool, boat, shelter, vessel).
- Latin binomials dropped.

A second Gemini pass applied to the first-pass output catches the long tail and additionally:

- Cultural-religious titles (sultan / padishah / shah / kadi / mullah / dervish / brahmin / bai / lama / raja / khan / tsar) $\rightarrow$ neutral “ruler” or “holy person” or “wealthy person”, explicitly *not* the Western-coded “king” / “queen” / “priest” / “monk”.
- Tradition-specific named characters (Baba Yaga, Kivioq, Coyote-as-character, Nach Ak Yum, Quetzalcoatl, Khevki, Mitoshka, etc.) $\rightarrow$ “character”.

Kept by design as cross-cultural anchors: cross-cultural religious figures (Christ, Buddha, Adam, Eve, Allah, God, Satan, named saints, Krishna, Rama, Sita), universal personifications (Sun, Moon, Wind, Sky, Earth, Sea, Thunder, Lightning, Star), all astronomical references (Pleiades, Orion, Big Dipper, Sirius, Venus, Milky Way, named constellations), all class words, cardinal directions, and generic universal vocabulary (forest, mountain, river, village, house, person, woman, man, child, day, night, fire, water). The narrative content of each motif, transformations, family relationships, magical events, plot, is preserved.

A held-out audit of 30 random LLM-anonymised motifs by an independent LLM judge finds a clean rate of 80–100% (with one strict batch reporting 100% and a more aggressive batch 60% because of borderline cultural items such as “prayer rug” or “tortillas”). After anonymisation, we apply a *model-predictability biome-tell filter* that operationalises “biome tell” as the only quantity that matters for our experiment: text-only recoverability of biome assignment by the embedding model that runs the headline test. For each LLM-cleaned English motif text M with assigned biome set B_M we compute its SigLIP-2 text embedding $\mathbf{t}_M$, form a visual prototype for each biome b as the mean of all biome- b iNaturalist image embeddings $\mathbf{p}_b$, score $s(M, b) = \cos(\mathbf{t}_M, \mathbf{p}_b)$, and define the tell $z(M) = [\max_{b \in B_M} s(M, b) - \overline{s(M, b')}] / \text{std}_{b' \notin B_M} s(M, b')$. We calibrate against a corpus-internal null in which the motif $\rightarrow$ biome assignment is scrambled 1,000 times across motifs (preserving the assignment-set-size distribution), set the threshold at the 95th percentile of the null z -distribution ($z_{95} = 2.83$), and drop motifs above it. The procedure removes 237 of 2,158 LLM-clean motifs (11.0%) and is independent of biome word lists: by construction, a kept motif’s biome assignment cannot be recovered from its anonymised text alone more reliably than chance text. The pipeline is *noun-targeted*: it strips species, place, biome, and cultural-artefact nouns and removes whole motifs that are text-recoverable to a biome by the embedding model, but does not strip verbs or activity predicates. A residual narrative such as “gathers fruit, steps in mammal dung” therefore retains its activity-level biome correlates (gathering fruit is more frequent in temperate and boreal forest descriptions than in desert ones). We return to this scope and the morphology-vs- activity-priming question in the Discussion; a follow-up activity-vocabulary biome-tell filter is the natural next anonymisation pass.

Within-taxon image-side stratification.

Because iNaturalist observations are taxonomically structured, some biomes are disproportionately rich in particular iconic taxa (boreal/tundra/conifer forests are 55–66% Plantae; mangroves only 11%). A motif whose anonymised text emphasises one taxon class will therefore score high in any biome whose iNat image pool emphasises the same taxon class, even when the biome’s specific visual character contributes nothing. To remove this biome-by-taxon co-occurrence as a possible explanation for the alignment, we recompute Δ_b stratified by the iNaturalist iconic-taxon label. For each biome b and taxon stratum t with at least 20 images, we compute the within-stratum $\Delta_{b,t} = \overline{S_{i,\mathcal{M}_b}}_{i \in b \cap t} - \overline{S_{i,\mathcal{M}_{\bar{b}}}}_{i \in b \cap t}$, and report the uniform across-strata mean as $\Delta_b^{\text{strat}} = (1/T_b) \sum_t \Delta_{b,t}$. The permutation null is the same scrambled-motif null applied within strata. The headline $\mu\Delta = +0.48 \times 10^{-3}$ and the seven robustly significant biomes (Tropical/Subtropical Moist Broadleaf, Tropical/Subtropical Dry Broadleaf, Tropical/Subtropical Grasslands & Savannas, Montane Grasslands & Shrublands) are reported under this stratification. Per-text-taxon decompositions (Figure 6) retain the marginal definition because stratifying on both motif-text taxon and image iconic taxon yields under-powered cells.

We separately explored simpler rule-based anonymisation pipelines (genus-level hypernymisation; a hand-curated class-level Russian hypernym map of roughly 3,500 entries; an audit-extended map of 6,300 entries) and found that the effect is positive under each of them and the across-biome ranking is preserved, ruling out the interpretation that any particular vocabulary anchor is doing the work. The LLM-mediated pipeline described above is the strictest in the family and the one we use for the headline.

Statistical tests

Image–motif similarities form an $n_{\text{img}} \times n_{\text{motif}}$ matrix $\mathbf{S}$. Per-motif residualisation (subtracting the motif grand mean) eliminates the contribution of generic motif valence to per-biome aggregates. For each biome b with n_b images and motif partition $(\mathcal{M}_b, \mathcal{M}_{\bar{b}})$, where Spec A defines $\mathcal{M}_b$ as the set of motifs touching ≤ 3 biomes and having ≥ 3 own-traditions in b , we report

$$\Delta_b = \frac{1}{n_b} \sum_{i \in \text{imgs}(b)} \left[\overline{S_{i,\mathcal{M}_b}} - \overline{S_{i,\mathcal{M}_{\bar{b}}}} \right],$$

with p -values from 1,000 permutations of the motif-to-biome assignment, and FDR control across biomes via the Benjamini–Hochberg procedure [Benjamini and Hochberg, 1995]. For all main-figure panels we apply a display threshold of `MIN_MOTIFS = 10`; biome cells with fewer than 10 in-biome motifs are shown in supplementary panels only. To bound the residual contribution of class-level anchors (the fact that after the LLM-mediated anonymisation the encoder still sees “mammal”/“bird”/“tree” rather than specific species), we additionally embed a variant in which every animal class word is further collapsed to a single token “animal”; this is the maximally aggressive control on animal vocabulary, preserving narrative grammar while removing every distinction between animal classes.

Data and code availability

All code, embeddings, the LLM-clean motif texts (Gemini-3.5-Flash output), the SigLIP-2 visual prototypes used by the model- predictability biome-tell filter, the held-out anonymisation audit, and the earlier rule-based hypernym maps (v3 through v8, released for completeness) are available at [repository URL placeholder]. The Berezkin abstracts are accessible at <http://www.ruthenia.ru/folklore/berezkin/>.

Acknowledgements

[Placeholders.]

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Supplementary materials

The supplementary materials below collect the analyses that probe the robustness of the headline finding from five complementary angles. We report (i) a per-taxon decomposition that asks which iNaturalist taxonomic groups carry the alignment, (ii) two anonymisation-progression ladders that follow the effect across five stages of progressively stricter text stripping, on iNaturalist and on Places365, (iii) a cross-model and cross-language replication across four vision-language model architectures and two text languages, (iv) a whole-corpus residualised panel without the Spec A breadth restriction, and (v) an analysis of the raw ecological composition of motif text per biome, which makes explicit what the species hypernymisation layer is removing. Throughout, the same statistical conventions apply as in the main text: per-cell residualised Δ with 1,000 motif-to-biome permutations and Benjamini-Hochberg FDR within each panel, the display threshold $\text{MIN_MOTIFS} = 10$, and gold marks for $q < 0.05$.

S1. Per-taxon decomposition (small-multiples view)

Figure 8 re-renders the per-taxon decomposition of the headline result as small-multiple bar panels, one taxon per panel, with within-taxon BH-FDR. This is a complementary view of the same data as Figure 6: rather than collapsing a taxon’s twelve biome cells into one violin, each panel shows the per-biome Δ values themselves, so the reader can see which specific biomes carry each taxon’s effect. The “all” panel reproduces the headline ranking (Tropical Grasslands, Mediterranean, Boreal/Taiga at the top); the Aves and Plantae panels echo this pattern; the Reptilia, Mollusca, and Arachnida panels show little structure, consistent with the per-taxon violin summary.

S3. Cross-model and cross-language replication

Figure 9 tests whether the across-biome ranking of Δ is an artefact of the specific vision-language model architecture used to produce the embeddings (SigLIP-2-large in the headline analysis). For this robustness check we re-run the residualised Spec A pipeline with four distinct vision-language model families across two text languages, using the earlier hand-curated class-level Russian hypernym map (v7) as the text anonymisation rather than the LLM-mediated pipeline used in the headline; the per-biome Δ scale therefore differs from the headline numbers but the across-biome ranking is the test of interest. The top row uses the Russian-language abstracts under SigLIP-2-large and under M-CLIP (XLM-Roberta-Large paired with CLIP-ViT-L/14). The middle and bottom rows use machine-translated English one-liner motif summaries under SigLIP-2-large, OpenCLIP-ViT-L/14 trained on LAION-2B, and OpenCLIP-ViT-L/14 trained on the OpenAI corpus. The scatter panels show per-biome Δ for matched model pairs. Spearman ρ lies in $[+0.71, +0.82]$ across cross-model pairs; the biome ranking is preserved across every model family tested. The across-biome ranking is therefore not a SigLIP-specific artefact.



Figure 8. Per-taxon biome panels. Per-taxon decomposition of the headline result, presented as small-multiple bar panels with within-taxon BH-FDR. Each panel shows the per-biome Δ for one taxon; biome row order is preserved across panels. Gold stars mark $\text{BH-}q < 0.05$ within the panel. Complements Figure 6.

S5. Raw ecological composition per biome

Figure 10 reports the raw ecological composition of motif text per biome, before anonymisation. For each WWF biome (rows) and each of fourteen taxonomic and botanical categories (columns), the left panel of the heatmap shows the percentage of Spec A motifs whose raw Russian abstract mentions at least one species or plant of that category. The right panel shows the difference between the Spec A subset and the all-in-biome pool: a positive cell means the category is over-represented in biome-specific motifs relative to the universal-dominated full pool. The heatmap therefore makes explicit, at the textual level, which taxonomic categories carry biome-specific signal before the hypernymisation layer removes their names from the encoder’s input. The figure helps motivate which species-vocabulary categories the v7 hypernym map needed to cover comprehensively, and clarifies the ecological content the encoder is no longer reading at the headline stage.

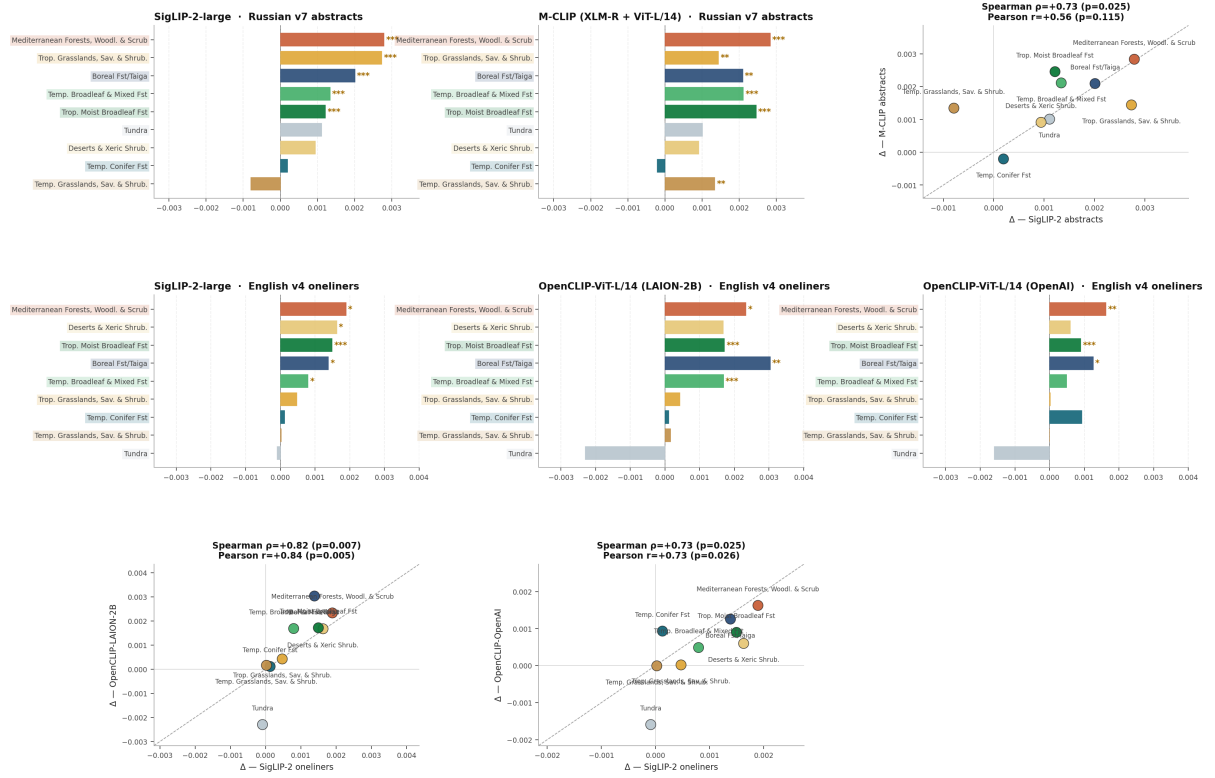


Figure 9. Cross-model robustness. *Top row:* Russian abstracts with SigLIP-2-large and M-CLIP (XLM-Roberta-Large + CLIP-ViT-L/14), plus their per-biome scatter. *Middle row:* English one-liners with SigLIP-2-large, OpenCLIP-ViT-L/14-LAION-2B, and OpenCLIP-ViT-L/14-OpenAI. *Bottom row:* per-biome scatters between SigLIP-2 oneliners and each OpenCLIP model. Spearman $\rho \in [+0.71, +0.82]$ across all cross-model pairs.

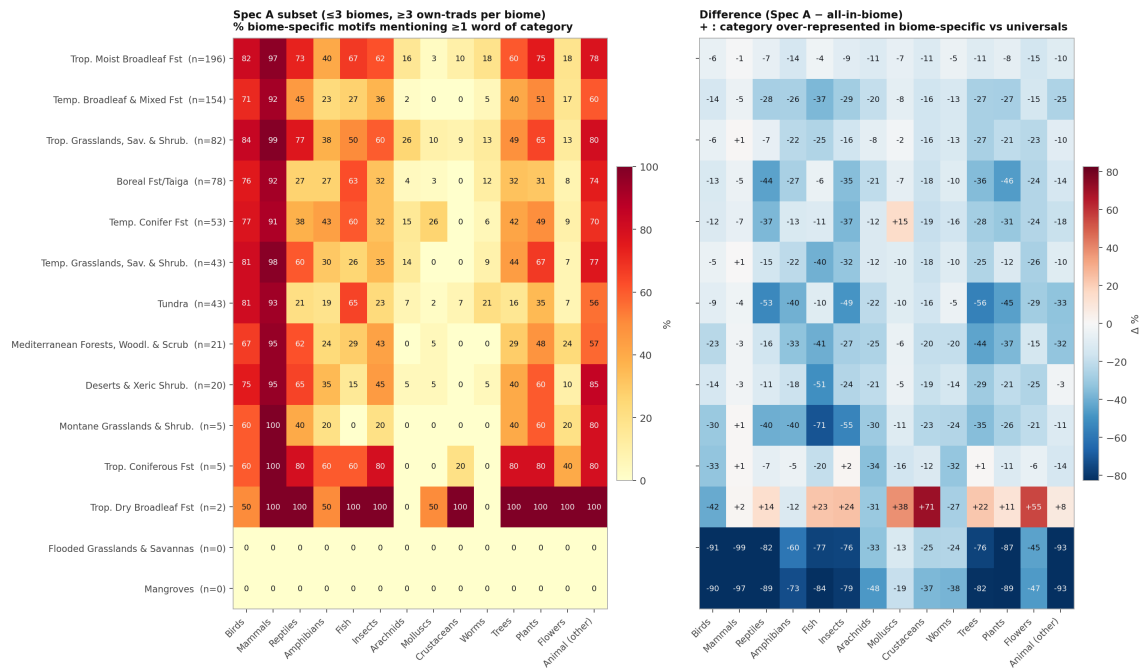


Figure 10. Ecological composition per biome (Spec A subset). *Left:* for each WWF biome (rows) and each of 14 taxonomic and botanical categories (columns), the cell shows the percentage of *Spec A motifs* (motifs touching ≤ 3 biomes, ≥ 3 own-traditions in that biome, the same selection that drives the headline result) whose raw Russian abstract mentions at least one species or plant of that category, before anonymisation. *Right:* difference (Spec A minus all-in-biome). Positive cells mark categories over-represented in biome-specific motifs relative to the universal-dominated full pool, revealing the ecological fingerprint that the species-hypernymisation layer subsequently removes. The all-motifs version of the heatmap is dominated by universal motifs and therefore uniformly red across biomes; we omit it here and release the underlying counts in the data drop.